

Andrew Seohwan Yu

Multimodal generative AI fine-tuning and inference optimization
3D medical image / video generation and understanding

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Education

Case Western Reserve University

- PhD Candidate in Computer Science
- Qualifying Exam: Evaluation of Image Generative Models
- GPA 3.35/4.00

Cleveland, OH

2021 -Present

Dec 2023

Cleveland State University

- Master of Computer and Information Sciences
- Thesis: NBA Basketball Analytics with ML
- *Magna Cum Laude*, GPA 3.61/4.00

Cleveland, OH

2014 - 2017

Experience

Graduate Research Assistant, Cleveland Clinic Research

- Advisor: Xiaojuan Li, PhD
- Multimodal generative AI: text, image, and 3D medical image generation
- Clinical applications: clinical report generation, radiology measurements

Cleveland, OH

2021 - Present

Graduate Research Assistant, Case Western Reserve University

- Advisor: Vipin Chaudhary, PhD
- Prompt engineering and optimization, and fine-tuning for narrow applications
- Anomaly detection: grounded distribution representation via embedding typicality

Cleveland, OH

2021 - Present

Awards

Cleveland Clinic Research, Catalyst Grant

- Co-Authors: Mingrui Yang, PhD, Pegah Ahadian, MS
- Radiology reporting system for clinical, diagnostic, and personalized care
- Multimodal generative AI, using real-time audio, 3D medical images, and reinforcement learning

Cleveland, OH

Jan 2026

Skills

Machine learning, deep learning, computer vision, MLOps, Python, PyTorch, PyTorch Lightning, Weight and Biases, scikit-learn, pandas, matplotlib, large language models, LLM, vision language models, VLM, generative adversarial networks, diffusion models, out-of-distribution detection, generative AI, Google Cloud Platform

Select Publications

[MICCAI 2026]: **Medical Image Spatial Grounding with Semantic Sampling**. Andrew Seohwan Yu, Mohsen Hariri, Kunion Nakamura, Mingrui Yang, Xiaojuan Li, Vipin Chaudhary. Medical Image Computing and Computer Assisted Intervention Society, Strasbourg, France, September 2026

[ICLR 2025]: **ForTe: Finding Outliers with Representation Typicality Estimation**. Debargha Ganguly, Warren Morningstar, Andrew Seohwan Yu, Vipin Chaudhary. International Conference on Learning Representations, Tampines, Singapore, April 2025

[MDPI 2025]: **Unsupervised Segmentation of Knee Bone Marrow Edema-like Lesions Using Conditional Generative Models**. Andrew Seohwan Yu, Mingrui Yang, William Holden, Ahmet Hakan Ok, Sameed Khan, Jeehun Kim, Carl Winalski, Naveen Subhas, Vipin Chaudhary, and Xiaojuan Li. Bioengineering 2024, 11, 526. May 22, 2024

Publications (Continued)

Inpainting MRI for unsupervised knee bone marrow edema-like lesion segmentation using conditional diffusion models, Andrew Seohwan Yu, Richard Lartey, William Holden, Ahmet Hakan Ok, Jeehun Kim, Carl Winalski, Naveen Subhas, Vipin Chaudhary, and Xiaojuan Li, presented at the Society of Photo-Optical Instrumentation Engineers (SPIE) Imaging Informatics for Healthcare, Research, and Applications, San Diego, February 20, 2024

Novel Unsupervised Segmentation of Bone Marrow Edema-Like Lesions using Bayesian Conditional Generative Adversarial Networks, Andrew Seohwan Yu, Sibaji Gaj, William Holden, Richard Lartey, Jeehun Kim, Carl Winalski, Naveen Subhas, and Xiaojuan Li, Proceedings of the International Society for Magnetic Resonance in Medicine, (ISMRM) Scientific Meeting and Exhibition, ISSN 1545-4428 (Online), May 19, 2023

Empirical Study: Temporal and Spatial Feature Processing Methods for Prediction of NBA Basketball Plays for Sports Analytics, Sun Sunnie Chung and Andrew Yu. Accepted to International Journal of Networked and Distributed Computing (IJNDC), Vol 7: Issue 3, ISSN Print: 2211-7938, ISSN Online: 2211-7946, July 2019

Automatic Identification and Analysis of Basketball Plays: NBA On-Ball Screens, Andrew Yu and Sun Sunnie Chung, in the Proceedings of the 4th IEEE International Conference on Big Data, Cloud Computing and Data Science Engineering, Honolulu, May 2019

Teaching

Pennsylvania State University

- Full-time instuctor
 - Artificial Intelligence (Python)
 - Technical Game Development (Unreal Engine 4)
 - Game Development Project (Unreal Engine 4)
 - Applications Programming (Android, Kotlin)
 - Operating Systems and Programming (C, UNIX)
 - Introduction to Programming Techniques (C++)

Erie, PA
2017 - 2021
Spring 2021
Spring 2021
Fall 2020
Spring 2020
Fall 2017 - Spring 2019
Fall 2017 - Summer 2021

Cleveland State University

- Graduate Teaching Assistant
 - Introduction to Engineering Design (C, Arduino)
 - Introduction to Programming (Java)

Cleveland, OH
2016 - 2017
Spring 2017
Fall 2016